Statistical Assessment of Operational, Financial and Service Delivery Performance among Water Service Providers in Kenya

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1 Introduction

1.1 Background of the Study

Water service providers play an important role in delivering safe and reliable water to households, businesses and institutions. Their performance can be assessed using operational, financial and service delivery indicators, including water production, billed water, non-revenue water, customer connections, turnover and compliance with performance standards.

Non-revenue water (NRW) refers to water supplied into a distribution system that does not generate corresponding revenue. It may arise from physical losses, apparent losses such as meter inaccuracies, and authorised consumption that is not billed. High NRW can affect the financial sustainability and operational efficiency of water service providers.

Statistical analysis provides a systematic way of describing performance, comparing providers, identifying relationships between indicators and investigating differences across providers. Such analysis can support evidence-based decisions aimed at improving operational efficiency and service delivery.

This study examines the operational, financial and service delivery performance of water service providers in Kenya using data for the 2024/2025 reporting year.

1.2 Statement of the Problem

Water service providers operate under different conditions and may experience substantial differences in operational efficiency, financial performance and service delivery. However, these differences need to be examined systematically to identify performance patterns and relationships among the available indicators.

In particular, non-revenue water may be associated with other operational and financial indicators, but the nature and strength of these relationships should be established empirically rather than assumed. Statistical analysis of the available provider-level data can help identify areas requiring further investigation and support evidence-based performance improvement.

1.3 General Objective

To statistically assess the operational, financial and service delivery performance of water service providers in Kenya during the 2024/2025 reporting year.

1.4 Specific Objectives

1. To describe and compare the operational, financial and service delivery performance of water service providers.

2. To examine the distribution of non-revenue water among water service providers.
3. To investigate relationships between non-revenue water and selected operational indicators.
4. To assess associations between financial performance and selected service delivery indicators.

1.5 Research Questions

1. How does performance vary among water service providers in Kenya?
2. What is the distribution of non-revenue water across the providers?
3. What relationships exist between non-revenue water and selected operational indicators?
4. How is financial performance associated with selected service delivery indicators?

2 Data Description and Methodology

2.1 Data Source

This study uses secondary data on water service providers in Kenya for the 2024/2025 reporting year. The dataset contains provider-level information on population coverage, customer connections, water production, billed water, non-revenue water, financial turnover, staffing and compliance status.

The data were obtained from the Excel workbook supplied for this study. The analysis treats each water service provider as an observational unit.

2.2 Study Design

The study adopts a quantitative, cross-sectional analytical design. It uses statistical techniques to summarize provider performance, compare indicators across providers and investigate associations among selected variables. Because the dataset represents a single reporting period, the analysis focuses on cross-sectional differences rather than changes over time.

2.3 Variables of Interest

The principal variables include:

- **Operational indicators:** water produced, billed water, active connections and total connections.
- **Financial indicator:** annual turnover, reported in millions of Kenyan shillings.
- **Service delivery indicators:** population served, population within the service area and compliance status.
- **Non-revenue water:** the reported percentage of water that does not generate corresponding revenue.

2.4 Statistical Methods

Descriptive statistics, including means, medians, standard deviations, ranges and frequency distributions, will be used to summarize the data. Tables and graphical displays will help identify differences and unusual observations across providers.

Correlation analysis will be used to examine associations between non-revenue water and selected numerical indicators. Compliance groups may also be compared where the data and assumptions support a meaningful comparison.

The choice of statistical method will depend on data quality, variable type, sample size and the assumptions of each method. Results will be interpreted as associations rather than causal effects.

2.5 Data Cleaning and Validation

The dataset will be checked for missing observations, duplicate provider names, invalid values and inconsistencies in reported indicators. Values that appear implausible will be flagged for investigation rather than automatically corrected. Missing or questionable values will be handled according to the requirements of each analysis, and the number of observations used will be reported where relevant.

3 Data Quality Assessment

3.1 Missing Values

The dataset was checked for missing observations before analysis. Table 1 summarises the missing values for each variable

Table 1: Missing Values by Variable

	Variable	Missing Observations	Missing (%)
provider	provider	1	1.05
population_service_area	population_service_area	1	1.05
population_served	population_served	1	1.05
total_connections	total_connections	1	1.05
active_connections	active_connections	7	7.37
towns_served	towns_served	4	4.21
turnover_ksh_million	turnover_ksh_million	1	1.05
water_produced_000_m3	water_produced_000_m3	1	1.05
domestic_kiosks_billed_000_m3	domestic_kiosks_billed_000_m3	1	1.05
total_billed_volume_000_m3	total_billed_volume_000_m3	1	1.05
non_revenue_water_pct	non_revenue_water_pct	4	4.21
production_per_capita_lcd	production_per_capita_lcd	18	18.95
consumption_per_capita_lcd	consumption_per_capita_lcd	1	1.05
total_staff	total_staff	1	1.05
compliance	compliance	1	1.05

3.2 Duplicate Provider Records

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No duplicate provider names were identified in the dataset.
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3.3 Implausible and Inconsistent Values

The dataset was also checked for negative active connection counts and unusual spacing in selected variables. Negative connection counts are not plausible as actual counts and require verification against the original source.

Table 2: Providers with Negative Active Connection Counts

provider	total_connections	active_connections
Kakamega Rural	5370	-3972
Tatu City	2802	-2802
Runda	1321	-1266

4 Statistical Results

4.1 Descriptive Analysis of Non-Revenue Water

Non-revenue water (NRW) is an important indicator of water-service-provider performance. This section summarizes the distribution of reported NRW percentages across the providers in the dataset.

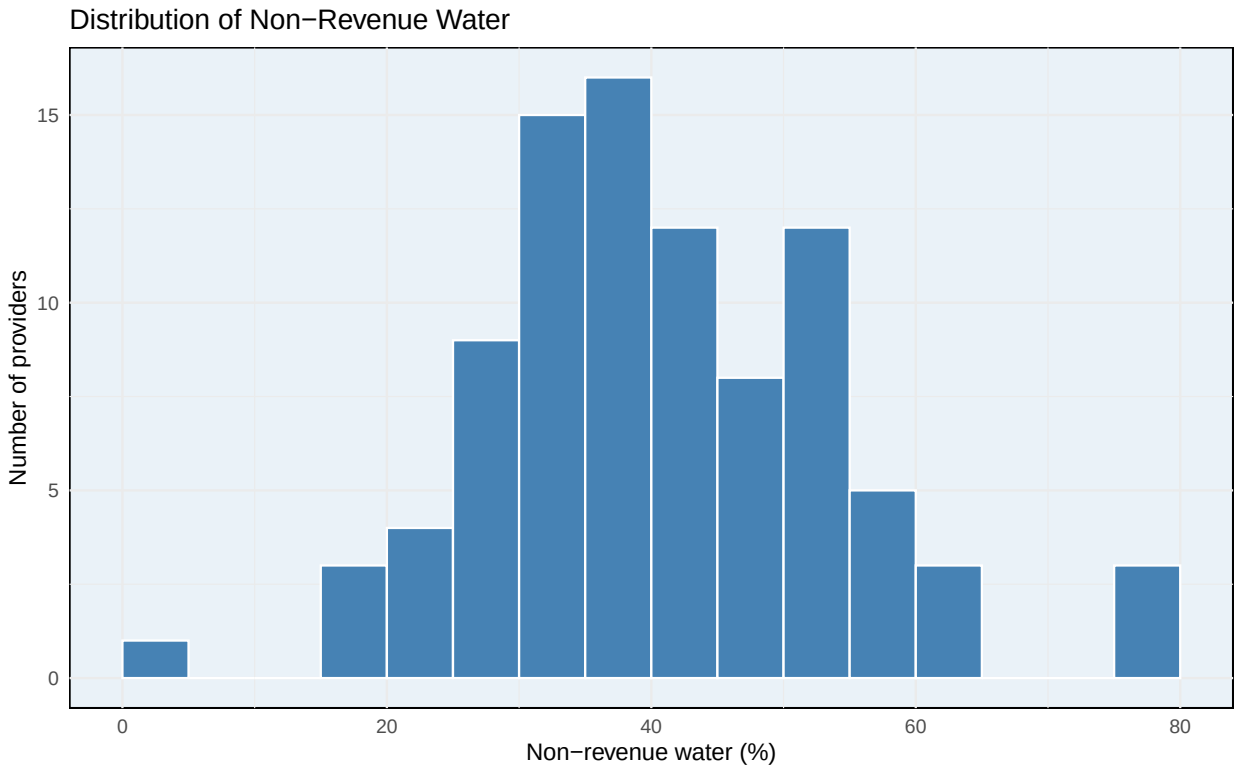
4.1.1 Summary Statistics

Table 3: Descriptive Statistics for Non-Revenue Water

Statistic	Value
Number of providers	91.00
Mean (%)	41.20
Median (%)	39.00
Standard deviation	13.37
Minimum (%)	2.00
Maximum (%)	80.00

The mean non-revenue water (NRW) was 41.20%, while the median was 39%. This indicates that reported NRW was substantial among the 91 providers with available observations. The standard deviation of 13.37 percentage points and the range of 2% to 80% indicate considerable variation in NRW across providers. These differences suggest that providers experienced varying levels of operational performance.

4.1.2 Distribution of Non-Revenue Water



Interpretation: Figure 1 presents the distribution of NRW percentages across the water service providers. The observed values range from 2% to 80%, indicating substantial variation in reported NRW. The histogram enables comparison of the frequency of providers across NRW levels.

4.2 Relationship between NRW and Operational Indicators

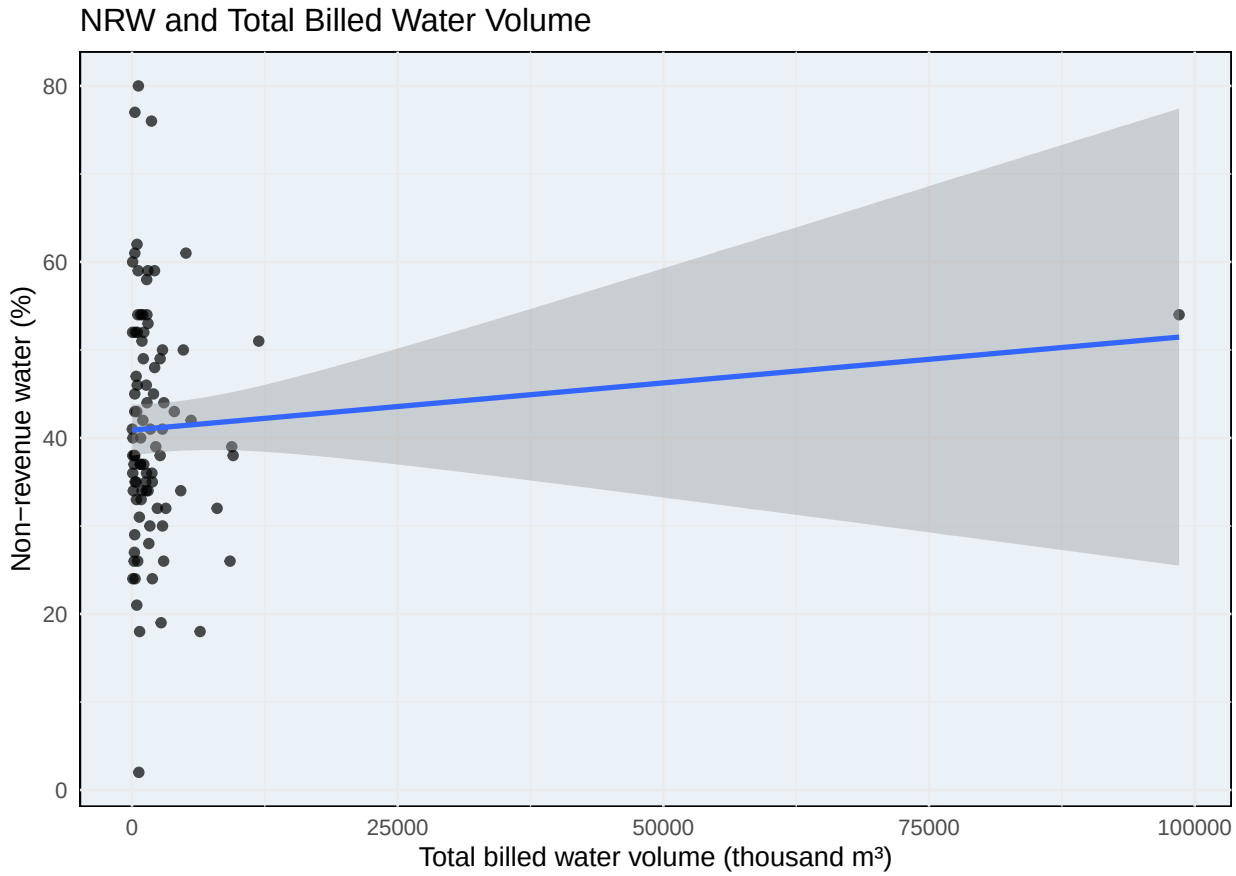
Pearson's correlation coefficients were used to examine the linear relationships between NRW and selected operational and financial indicators. Coefficients close to zero indicate weak linear associations, while positive coefficients indicate relationships in the same direction. The analysis uses available paired observations for each variable. Correlation does not establish causation.

Table 4: Correlations between NRW and Selected Indicators

Indicator	Paired Observations	Pearson Correlation
water_produced_000_m3	91	0.119
total_billed_volume_000_m3	91	0.083
total_connections	91	0.109
turnover_ksh_million	91	0.090

Interpretation: All four indicators had very weak positive correlations with NRW. The coefficients were 0.119 for water produced, 0.083 for total billed water volume, 0.109 for total connections and 0.090 for turnover. These results indicate that the indicators had little linear association with NRW in this dataset. The correlations do not establish cause-and-effect relationships.

4.2.1 Scatter plot: NRW and billed water volume



Interpretation: The relationship between NRW and total billed water volume was very weakly positive ($r = 0.083$). Therefore, the volume of billed water had little linear association with NRW among the providers studied. The scatterplot also allows to observe the spread of providers across different NRW levels.

4.3 Comparison of NRW by Compliance Status

This section compares reported non-revenue water percentages between providers classified as compliant and non-compliant. Group summaries are used to describe differences, while the Wilcoxon rank-sum test assesses whether the observed distributions differ statistically.

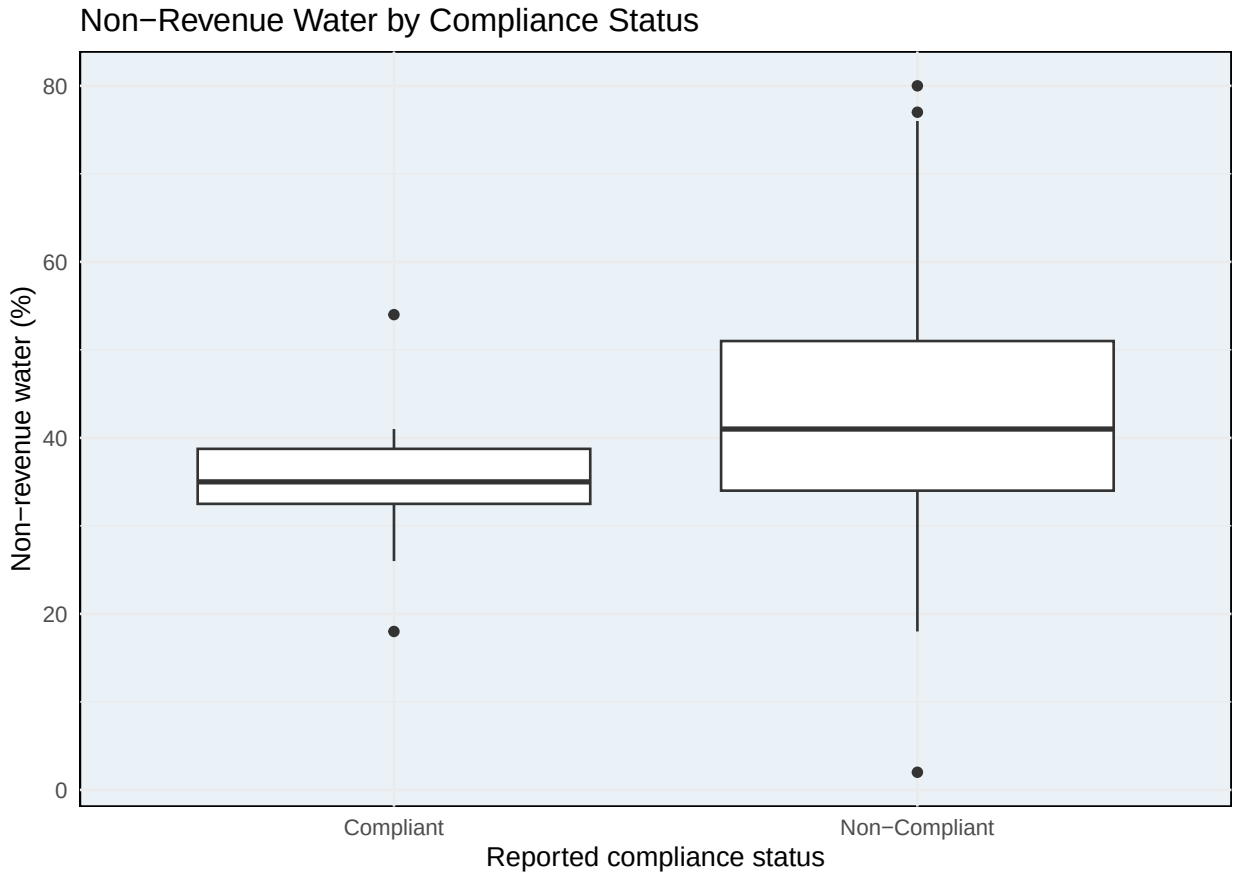
Table 5: NRW Summary by Compliance Status

Compliance Status	Providers	Mean NRW (%)	Median NRW (%)	SD (%)
Compliant	10	35.20	35	9.45
Non-Compliant	81	41.94	41	13.64

Interpretation: Non-compliant providers recorded a higher mean NRW of 41.94% compared with 35.20% among compliant providers. Their median NRW was also higher, at 41% compared with

35%. This suggests a descriptive difference between the groups, although the group sizes were unequal: 81 non-compliant providers and 10 compliant providers.

4.3.1 Visual Comparison



4.3.2 Statistical Test

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The test does not provide sufficient evidence of a  difference between the NRW distributions a
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Interpretation: The Wilcoxon rank-sum test did not provide sufficient statistical evidence of a difference between the NRW distributions of compliant and non-compliant providers at the 5% significance level. Although the non-compliant group had higher mean and median NRW values, the observed difference was not statistically significant according to this test.

5 Discussion of Findings

The study examined non-revenue water (NRW) and its relationships with selected operational and financial indicators among water service providers in Kenya during the 2024/2025 reporting year.

The descriptive analysis showed a mean NRW of 41.20% and a median of 39% among 91 providers. The observed range of 2% to 80% indicates considerable differences in NRW across providers. These findings suggest that NRW management is an important area for operational improvement, although the reasons for the differences cannot be established from the descriptive statistics alone.

Correlation analysis revealed very weak positive linear relationships between NRW and water produced, total billed water volume, total connections and turnover. The coefficients ranged from 0.083 to 0.119, indicating that none of these indicators had a strong linear association with NRW in the available data. Other factors not examined in these comparisons may also be relevant.

The compliance comparison showed that non-compliant providers had higher mean and median NRW than compliant providers. However, the Wilcoxon rank-sum test did not provide sufficient evidence of a statistically significant difference between the two distributions at the 5% significance level. The observed descriptive differences should therefore be interpreted cautiously.

The data-quality assessment also identified negative active connection counts for three providers. These values require verification because they are not plausible as actual connection counts. Such inconsistencies highlight the importance of validating administrative data before using them to support operational decisions.

6 Conclusion

The study found substantial variation in reported non-revenue water among water service providers in Kenya during the 2024/2025 reporting year. The mean NRW was 41.20%, while individual values ranged from 2% to 80%.

NRW had very weak positive linear relationships with water production, billed water volume, total connections and turnover. Although non-compliant providers recorded higher mean and median NRW than compliant providers, the statistical test did not establish a significant difference between their NRW distributions.

Overall, the findings demonstrate the value of statistical analysis in describing differences in water-provider performance. However, data inconsistencies and the limited scope of the analyses mean that the results should be interpreted cautiously. Further investigation using validated data and additional operational indicators could help explain the factors associated with NRW.

7 Recommendations

1. **Improve NRW monitoring:** Water service providers should regularly monitor NRW and investigate unusually high reported percentages.
2. **Strengthen data validation:** Reported connection counts and other operational indicators should be checked for negative, missing or inconsistent values before analysis.
3. **Investigate possible contributing factors:** Further analysis should examine additional operational indicators, including network condition, metering accuracy and billing practices, where reliable data are available.
4. **Improve compliance monitoring:** Providers should use routine performance assessments to identify areas requiring corrective action, without assuming that compliance status alone explains NRW differences.

5. **Expand future research:** Future studies could use data from several reporting years to investigate changes in NRW over time and assess whether operational improvements are associated with changes in performance.